

Recoverable but Unrouted: Auditing Tool-Call Routing in a Persistent Language-Model Agent*

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Abstract

A tool-using language-model agent can generate a parser-recoverable action and still do nothing: the command may appear in ordinary assistant text while the executor reads only the structured tool-call field. We study this silent action-routing failure in a 2B agent controlling a persistent game through 17 typed tools. A forensic audit first showed that historical score changes could not identify a training effect because model weights, visited states, rendered interfaces, and runtime recovery had changed together. We therefore moved from long-horizon scores to prospectively frozen manipulation checks. An 18-cell weights-by-recovery diagnostic produced no recovery-eligible output in 958 generations, so the recovery intervention never engaged. A fixed-state V1 study then found that presenting a registered tool schema increased recoverable calls stranded in text by 15–30 percentage points across three checkpoints, but a post-outcome audit found that its five nominal seeds were ineffective. We repaired the sampling path and prospectively froze V2 on 20 outcome-unseen states before endpoint access. All 1,200 requests succeeded; all 240 state-condition groups contained five distinct semantic responses, and 126 groups varied in the primary outcome. The native-schema contrast was positive at Base, round 2, and round 3 (+23, +13, and +10 points), satisfying the registered directional criterion. A clearly labeled post-hoc analysis also found that schema-invalid structured envelopes occurred less often per request when the native schema was present. Because schema exposure also changed routing, this comparison is descriptive. The contribution is an artifact-centered audit of where actions are lost between model and environment, not a checkpoint ranking, recovery benefit, or established training effect.

1 Introduction

Tool use is often reported as though it were a property of model weights. In a deployed agent, however, an action must survive a chain of contracts. The model sees a rendered description of available tools, emits both text and structured fields, an endpoint serializes those fields, a parser accepts or rejects them, and a harness may rewrite or recover content before an environment changes state. A valid command at one layer can be invisible at the next.

This distinction matters most in long-horizon settings. If a command is silently dropped, the agent receives no state transition, later observations diverge, and an eventual task score attributes the failure to the entire policy. If model weights, prompts, schemas, parsers, and recovery rules also change between runs, the score does not reveal which component helped or failed. The resulting ambiguity is not fixed by collecting more turns from the same run.

We encountered this problem while auditing a small language-model agent in a persistent MMORPG. The agent receives a procedural quest guide and acts through 17 typed tools. Historical development records suggested that changing the training-state distribution improved quest progress, while later runs also changed

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the tool interface and runtime recovery. The records were rich enough to recover exact descriptive scores, but not to reconstruct every checkpoint, reset, renderer, or random seed. Rather than promote those scores into a causal claim, we used them to localize a testable interface mechanism.

The resulting study follows a sequence that is useful beyond this game. First, we content-bound the historical evidence and marked which comparisons were not identifiable. Second, we ran a prospective recovery factorial and checked whether the intervention actually fired. It did not: none of 958 generations was eligible. Third, a registered fixed-state grid isolated two model-visible interface factors and exposed recoverable calls that were printed in text rather than delivered through the structured channel. Fourth, an audit discovered that nominal request seeds had not changed sampling. We repaired the seed path and prospectively froze an outcome-unseen replication instead of retrospectively upgrading the first study.

Our contributions are:

1. We decompose tool execution into a model-to-environment routing contract and define auditable outcomes that distinguish a structured envelope, an recoverable call stranded in text, a malformed surface form, and no recoverable call.
2. We report a registered outcome-unseen finite-grid replication across three fixed checkpoints. Native schema exposure increased content-only recoverable-call incidence by 10–23 percentage points at every checkpoint, satisfying the prospectively frozen directional criterion.
3. We provide a failure-aware evidence sequence: content-bound historical recovery, a failed manipulation check, explicit discovery/replication separation, model-level seed validation, and independently recomputable raw artifacts.

The bounded conclusion is about this action-routing contract. The three checkpoints form one model lineage, the states come from one historical run, and the renderer is fixed. We do not treat checkpoints as training replicates, infer a recovery benefit, or claim improved quest performance.

2 Setting and Action-Routing Contract

2.1 Persistent tool-use setting

The environment is a permissioned fork of the open-source Kaetram MMORPG (Kaetram contributors, 2026). A Qwen/Qwen3.5-2B agent (Qwen Team, 2026) controls a persistent player through 17 tools for observation, navigation, combat, gathering, dialogue, quest inspection, inventory use, and crafting. Game state persists across conversation rollovers. The system prompt contains procedural quest guidance, so the benchmark measures execution of a supplied plan rather than autonomous discovery of game rules.

The historical development score, CORE-3, counts newly reached stages in three quest chains, with ten stages per agent and three prompt variants for a maximum of 30. This score is useful for end-to-end development, much like persistent state metrics in τ -bench (Yao et al., 2025), but it is too coarse to localize a dropped action. The registered interface studies therefore operate on fixed historical decision states and inspect the immediate model response.

2.2 How a recoverable action can disappear

Figure 1 separates the layers between a model-visible tool and an environment transition. The registered “native schema” treatment means that a hash-bound 17-tool schema is supplied through the model endpoint’s native `tools` field. It is not claimed to be byte-identical to every historical production schema.

Response route	Measured routing boundary	Beyond V2
Structured field	schema check → candidate action	execution / environment
Ordinary content	recovery parser → candidate action	execution / environment

Figure 1: Two response routes converge on a candidate action. V2 measures whether a candidate is stranded in ordinary content rather than delivered in the structured field; execution, environment acceptance, state change, and task utility are outside its primary outcome.

For example, a response may contain a canonical function block naming `navigate` with valid JSON-like arguments, while its structured `tool_calls` field is empty. The harness sees ordinary text unless an explicit recovery layer parses that block. We use the following taxonomy:

Structured envelope. The endpoint returns at least one call in the structured field. Schema validity is a separate question.

Content-only recoverable call. The structured field is empty, but the prospectively frozen deterministic parser recovers at least one candidate from the text. This is the primary outcome, also called a recovery opportunity.

No recoverable call. Neither route yields a candidate.

Malformed surface form. A secondary detector identifies a registered syntax family. This label is not synonymous with content-only: most recovered V1 calls were canonical function blocks printed into the wrong channel.

The first three categories are mutually exclusive for the primary analysis. The malformed-family label is secondary. A later post-hoc analysis additionally checks whether structured envelopes name a registered function and satisfy its argument schema; it does not redefine the primary outcome.

2.3 Interface factors and fixed checkpoints

Each registered grid crosses two model-visible factors. *Documentation* uses either historical Python-like descriptions or canonicalized prose. *Schema exposure* omits or includes the registered 17-tool schema through the native field. The four conditions are therefore Python/no-schema, Python/schema, canonical/no-schema, and canonical/schema.

We evaluate Base, OPD round 2, and OPD round 3 weight snapshots. These are fixed checkpoints chosen before outcome inspection, not independently trained draws. Every contrast is reported separately by checkpoint. A consistent direction across them is robustness within this lineage, not population-level evidence about training.

3 Evidence Protocol

Persistent agents need provenance not only for code and weights, but also for the state shown to the model, rendered prompt, tool schema, response channel, parser, and any history rewrite. We therefore grade an artifact before using it to support a claim. Table 1 summarizes the resulting evidence ladder.

Table 1: Evidence ladder and permitted inference. “V1” and “V2” are separate registered studies.

Evidence	Audit status	Permitted inference
Historical OPD runs	Scores replayed; launch identities incomplete	Descriptive case and hypothesis generation only
30-minute recovery grid	18/18 cells; 958 raw generations	Manipulation failed because no output was eligible
V1 interface grid	1,200/1,200; complete public rows	Fixed-panel discovery; nominal seeds ineffective
V1 seed audit	Semantic duplicates recomputed post hoc	Collapse to 20 outputs per cell; no sampling claim
V2 interface grid	1,200/1,200; outcome-unseen panel; effective gates	Registered finite-grid directional replication
Structured validity	Recomputed after V2 outcomes	Post-hoc routing diagnostic, not a new primary outcome

For the historical long-horizon runs, a fresh training seed would be the unit for a training-method claim, with evaluation worlds nested within it. Prompt variants sharing weights and a launch are not independent policies. For the interface studies, the estimand is explicitly the registered finite grid of state-seed pairs. States from one historical run may be temporally dependent, so we report exact paired finite-grid contrasts rather than p-values or population confidence intervals.

Here, “registered” means prospectively frozen in a clean commit before the selected states were sent to an endpoint. The design was not deposited in an external public registry; the claim is about prospective outcome separation, not public timestamping.

The V2 release contains the registration, selected rendered states, endpoint attestations, seed gates, all 1,200 raw rows, sealed analysis, code-closure hashes, and content indexes. A producer verifier replays the analysis; a secondary verifier reimplements the primary binary parser-recoverability label and recomputes cells, contrasts, seed heterogeneity, and the directional verdict. Exact recovered-call detail additionally uses the shared frozen canonicalization code. The artifact-index digest printed in this PDF binds the paper to the exact supplementary bundle. These checks establish artifact integrity and deterministic recomputation, not generalization.

4 Historical Case: Why Scores Did Not Identify the Lever

The historical sequence motivated the audit but is not its causal result. A natural-visitation OPD round changed aggregate tool-use behavior while remaining at 12/30. A later round mixed natural trajectories with rollouts begun from a loadable, walkability-checked intermediate player state and was followed by a 15/30 canonical-start run. Every prompt variant crossed one previously blocked quest stage. The round also changed initialization and data, and its reset and gameplay seed were not preserved. The defensible observation is therefore that a state-augmented training round was followed by wall passage in one run, not that state restoration caused it.

A later development campaign supplied nine mechanism arms. Content-bound replay recovers the reported six-hour totals 12, 18, 12, 15, 13, 14, 12, 17, and 14. Four historically intended corpus/init-matched comparisons replace teacher-derived token advantages with a positive constant; the constant-weight arm matches or exceeds the graded arm in three of four cells. A repaired-render graded sensitivity reverses one direction. Each cell is one historical training run, and the launch-time corpus, checkpoint, render, reset, and random-seed identities were not all sealed. These observations motivate a corpus and interface hypothesis but cannot estimate a grading or state-visitation effect.

The recovery affordance was also historically confounded with weights. A June round-3 run used recovery and scored 18/30; a recovered run labeled with round-3 weights and recovery disabled also replays to 18/30. Missing launch parity prevents a weights-only interpretation. Historical notes report recovery markers in

rewritten sessions, but raw pre-rewrite emissions were unavailable, so those markers cannot measure the natural trigger rate.

One separate evaluation lane was found to reset a different database from the one used by its game servers and was quarantined. The headline runs came from a different database-aligned orchestrator, and their recovered first observations match the canonical player start. Their exact historical reset command and game revision remain unattested. This distinction matters: finding one concrete infrastructure defect does not justify invalidating unrelated runs, but neither does it repair their missing provenance.

5 Prospectively Frozen Interface Studies

5.1 The recovery factorial failed its manipulation check

We first prospectively froze 18 local cells crossing three fixed checkpoints, recovery off/on, and three paired 30-minute seed blocks. All cells completed and 1,266 retained files rehashed. Across 958 raw generations, 294 contained a canonical structured call and 664 contained none. No generation matched the registered recovery eligibility rule, and no call was recovered. The recovery switch therefore never applied its intervention. Quest-stage differences among cells cannot be called recovery effects.

This null trigger count changed the next question. Instead of extending a factorial whose treatment did not engage, we asked which model-visible interfaces naturally produce content that the recovery layer could act on.

5.2 V1 discovery and the ineffective-seed audit

V1 crossed the three checkpoints, four interfaces, 20 historical decision states, and five nominal request seeds. All 1,200 requests completed. After outcomes, an independent diversity audit found one semantic response in every five-seed state-condition group: the serving thread accepted the request seed but did not change its sampling stream. We therefore collapse V1 to one output per state-condition group, or 20 outputs per cell. No stochastic-sampling uncertainty is claimed.

On that collapsed discovery panel, native schema exposure increased content-only recoverable-call incidence by 30, 22.5, and 15 percentage points for Base, round 2, and round 3. Documentation effects were mixed. These values motivated a directional replication criterion, but the ineffective seeds and outcome-seen state panel prevent V1 from serving as that replication.

5.3 V2 prospectively frozen outcome-unseen replication

V2 was frozen before any selected state was sent to an endpoint. It selected 20 different source logs from the same historical Base run, explicitly excluding all V1 source paths. Selection used registered path, metadata, and parseability rules rather than interface outcomes. The design remained $3 \times 4 \times 20 \times 5 = 1,200$ requests.

Before a checkpoint could create an outcome directory, an unrelated plain-English seed gate required five seeds to produce at least three semantic responses, while an exact repeat of the third seed had to reproduce exactly. Base, round 2, and round 3 all passed. The registered directional criterion was strict: the native-schema contrast had to be positive at all three checkpoints on the complete grid. Documentation and interaction contrasts had no directional success rule. Failures would remain in the artifact, and no verdict would be emitted unless every scheduled request succeeded.

The primary contrast averages schema-present minus schema-absent incidence over the two documentation conditions within each checkpoint. Documentation and interaction contrasts are computed analogously. We also report, for each paired state, whether the effect is positive, negative, or zero, plus semantic and primary-outcome heterogeneity across the five seeds.

5.4 V2 results

All 1,200 requests succeeded. Every one of the 240 checkpoint-condition-state groups contained five distinct semantic responses, showing that the repaired seed path changed generation on the study prompts as well as on the gate. The primary recovery-opportunity label varied across seeds in 126 of 240 groups. V2 therefore supplies non-identical within-state responses rather than deterministic replay; it does not make them independent draws from a deployment population.

Table 2: Recovery-opportunity incidence on the registered 20-state finite grid. Cells show opportunities/successful requests (rate). Contrasts are paired rate differences in percentage points; +/−/0 counts the states with positive, negative, or zero effects. Descriptive fixed-grid results; 126/240 state-condition groups vary in the primary outcome across five effective paired seeds, and the registered all-checkpoint direction criterion is met.

Weights	Python / none	Python / native	Canonical / none	Canonical / native
Base	17/100 (17.0%)	39/100 (39.0%)	13/100 (13.0%)	37/100 (37.0%)
Round 2	16/100 (16.0%)	27/100 (27.0%)	15/100 (15.0%)	30/100 (30.0%)
Round 3	26/100 (26.0%)	36/100 (36.0%)	21/100 (21.0%)	31/100 (31.0%)

Weights	Native schema	Canonical docs	Interaction
Base	+23.0 (14/1/5)	-3.0 (5/7/8)	+2.0 (6/5/9)
Round 2	+13.0 (12/2/6)	+1.0 (7/5/8)	+4.0 (7/4/9)
Round 3	+10.0 (13/3/4)	-5.0 (3/7/10)	+0.0 (6/8/6)

Native schema exposure increased the registered primary outcome by 23 points at Base, 13 at round 2, and 10 at round 3. All three effects were strictly positive, so the prospectively frozen directional criterion passed. The magnitudes are smaller than V1’s 30, 22.5, and 15 points, which is expected to matter when interpreting the discovery study. Canonicalized documentation had small mixed effects (−3, +1, and −5 points); the interactions were +2, +4, and 0 points. We do not pool checkpoints or attach population uncertainty to the twenty states.

5.5 Post-hoc schema validity of structured envelopes

The prospectively frozen primary outcome asks whether a recoverable candidate was stranded in text; it treats any nonempty structured field as “structured.” After V2, we asked a different question: do structured envelopes name a registered tool and satisfy its argument schema? Table 3 reports this explicitly post-hoc analysis. Counts pool the two documentation conditions within a checkpoint and show invalid structured envelopes per 200 requests, without conditioning on whether a structured envelope was emitted.

Table 3: Post-hoc counts of schema-invalid structured envelopes per 200 requests. This descriptive analysis does not redefine the registered primary outcome.

Weights	Schema absent	Schema present
Base	34	1
Round 2	24	6
Round 3	18	3

In this post-hoc descriptive analysis, schema-invalid structured envelopes occurred less often per 200 requests when the native schema was present, even while recoverable calls stranded in text became more common. Because schema exposure also changed whether responses reached the structured channel, comparisons conditional on producing a structured envelope are selection-sensitive and are not interpreted causally. The registered study does not determine whether recovering and executing stranded calls improves environment state, nor does the post-hoc table define a net-utility metric.

6 What the Audit Establishes

A manipulation check precedes an effect estimate. The recovery factorial completed cleanly, but completion was not evidence that recovery had been tested. Because eligibility was zero, the treatment contrast was undefined in practice. Trigger incidence had to be measured first.

Model-visible tools and executor-visible calls are different. Providing a schema changes the distribution over both output content and the structured channel. A schema-parity test that only hashes the tool list does not establish parser alignment; raw response fields and parser outcomes must also be retained.

A structured envelope is not automatically executable. The post-hoc audit found unknown functions, unknown or missing arguments, and wrong argument types. Evaluations should report schema validity after channel routing rather than equating a nonempty envelope with a valid action.

Request parameters are not effective randomization by inspection. V1 stored five distinct seed values, yet they produced identical semantic outputs. V2 required model-level diversity plus exact repeatability before outcome collection and audited heterogeneity after completion.

Artifact integrity does not create experimental independence. Cryptographic manifests establish which bytes were analyzed. They do not turn three related checkpoints into training replicates or twenty temporally related states into an independent population sample. The statistical claim must remain conditional on the registered finite grid.

Together, these lessons suggest a reusable audit order for tool agents: bind the interface, retain raw channels, verify the treatment fires, test the effective randomization path, classify calls at each routing stage, and only then measure downstream task effects.

7 Related Work

Function-calling benchmarks increasingly distinguish structured-call validity from broader agent behavior. BFCL evaluates function selection and formatting, including stateful and multi-step settings (Patil et al., 2025). ToolPRM studies fine-grained structured generation and the cost of early JSON errors (Lin et al., 2026); GraphQLRestBench makes response schemas visible for sequential API calls (Saha et al., 2024). Meanwhile, τ -bench emphasizes persistent end states and repeated-trial reliability (Yao et al., 2025). BAL-ROG and Orak extend game-agent evaluation across multiple environments and interfaces (Paglieri et al., 2025; Park et al., 2025). Our study is narrower: it traces one action through a renderer, response channel, schema checker, parser, and recovery layer. It does not propose MCP or typed tools as novel infrastructure.

Recovery can also be learned or implemented at different layers. Fission-GRPO learns from execution errors (Zhang et al., 2026); our recovery switch is a runtime rewrite for a fixed checkpoint and cannot be described as learned self-correction. The distinction mirrors the central theme of this paper: weights, rendered affordances, parser behavior, and harness intervention are separate treatments.

The historical motivation comes from on-policy distillation. GKD trains on student-generated sequences (Agarwal et al., 2024), and later work changes state occupancy through teacher-success prefixes, mixed teacher/student turns, prefix replay, or first-error curricula (Wang et al., 2026; Li et al., 2026; Liao et al., 2026; Lyu et al., 2025). Dagger gives the complementary imitation-learning lesson that training data must cover learner-induced states (Ross et al., 2011); Backplay and Go-Explore use intermediate resets to expose sparse reward (Resnick et al., 2018; Ecoffet et al., 2021). These works preempt a broad novelty claim for intermediate starts. Our historical state-restoration hypothesis remains prospective and is included in Appendix A only to document what the routing audit does and does not establish.

Finally, reproducibility checklists provide a general baseline (Pineau et al., 2021). Persistent tool agents add bindings among world state, visible history, render contract, response channels, parser, and history rewrites. The evidence ladder here operationalizes those additional bindings for one case.

8 Limitations

V2 is a registered finite-grid study, not an independent population sample. Its twenty outcome-unseen states were selected from different source logs than V1 but from the same historical Base run, and temporal dependence may remain. The three checkpoints are related members of one model lineage and are not training replicates. All requests use one model family, one registered 17-tool schema, one renderer, and one deterministic recovery parser. A second model family, renderer, parser, state pool, and environment are needed before a broad interface claim.

The primary outcome stops at parser recoverability. It does not show that a recovered call is safe, semantically appropriate, accepted by the game server, or beneficial for quest progress. Conversely, a no-call response can be appropriate when deliberation is useful. A controlled execution study should separately measure parser acceptance, environment acceptance, immediate state change, and downstream utility.

V1’s five nominal seeds were ineffective and were collapsed after a post-outcome audit. V2 repairs this specific failure: all 240 groups have five unique semantic responses, and 126 vary in the primary label. That does not make the five samples independent draws from a stable deployment distribution, and no p-values or confidence intervals are reported.

The structured-call validity analysis was conceived after V2 outcomes. It is reported because it prevents the misleading equation of “structured” with “executable,” but its unconditional counts are exploratory. Schema exposure also affects channel selection, so validity conditional on entering the structured channel is selection-sensitive and receives no causal interpretation. The analysis does not change the registered primary outcome or directional verdict.

Historical long-horizon scores remain descriptive. Some raw transcripts are private, launch-time checkpoint/render/reset identities are incomplete, and one separate database-misaligned lane was quarantined. The public historical score projection permits exact score replay but not independent repetition of the private raw-to-projection extraction. The prompt also contains procedural quest knowledge, so the game benchmark measures plan execution rather than autonomous world-model acquisition.

Broader impact. The experiments run in an isolated game environment and use no human-user data. The main risk is interpretive: a routing diagnostic can be mistaken for evidence of agent competence, safety, or downstream utility. We therefore release raw channels and stage-specific outcomes and explicitly stop the primary claim before execution or environment change.

9 Conclusion

An agent can generate a parser-recoverable action that never becomes an environment action. In our fixed-state study, native schema exposure consistently increased such content-only calls across three fixed checkpoints: +23, +13, and +10 percentage points on a prospectively frozen outcome-unseen panel. All 1,200 requests succeeded, effective seed diversity was verified, and the all-checkpoint direction criterion passed. In a post-hoc descriptive audit, schema-invalid structured envelopes also occurred less often per request when the native schema was present; because routing changed too, this is not a conditional causal validity effect.

These findings do not rank checkpoints or establish a recovery or training benefit. They show why tool-agent evaluation must treat weights, rendered interface, response channel, parser, and runtime repair as distinct components. The broader methodological lesson is simple: before attributing a long-horizon score to a model, verify that the intended action could travel through the entire system.

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A Prospective State-Visitation Hypothesis

The historical case began with a state-visitation hypothesis for on-policy distillation. Let x be persistent player state, h the model-visible history, and $d_{\mu,q}^{\pi}(x,h)$ the joint occupancy induced by policy $\pi(a|h)$ from player-state initializer μ and history constructor $q(h|x)$. A reverse-KL on-policy objective is

$$\mathcal{L}_{\text{OPD}} = \mathbb{E}_{(x,h) \sim d_{\mu,q}^{\pi_S}} \mathbb{E}_{a \sim \pi_S(\cdot|h)} [\log \pi_S(a|h) - \log \pi_T(a|h)]. \quad (1)$$

If teacher advantage is concentrated on states with negligible student occupancy, natural student rollouts provide little supervision there.

The prospective intervention restores witness-certified persistent player states selected by a frozen visitation, teacher-advantage, recoverability, and task-relevance rule. The implementation restores player state, not the complete shared world. Because a database snapshot is not the policy state, the planned study crosses each restored x with minimal canonical, authentic-prefix, and matched reconstructed histories. It compares natural OPD, targeted states, random-valid and progress-matched resets, teacher-success prefixes, and turn-level guidance under matched budgets and independent training seeds.

No verified trained-arm outcome exists. This protocol is a hypothesis generated by the historical audit, not a contribution established by the V1 or V2 interface studies.

B Historical Training Details

The following settings are reconstructed from implementation and development notes rather than a complete launch-time receipt. A scaffolded Qwen/Qwen3.5-4B teacher scored tokens emitted by a 2B student. Training used a clipped importance-weighted update, one epoch, learning rate 5×10^{-5} , effective batch size 32, and LoRA initialized over the prior merged checkpoint. Action positions received the language-model loss.

Round 1 contains 5,564 training and 574 held-out records from natural Base rollouts. Round 2 combines 5,523 natural round-1 records with 2,326 records collected after loading a player at an accepted Herbalist quest, Foraging level 5, starter inventory, and a walkability-checked tile near the target resource. Evaluation did not intentionally load that state. The historical split held out every tenth session in deterministic file order and lacks the stronger run-stratified, duplicate-aware provenance required for a new causal study.

C Claim-to-Evidence Boundary

Table 4: Claims supported by the current evidence.

Claim	Boundary
Historical score replay	Nine July scores are content-bound descriptive observations; treatment identities and effects are not established.
Recovery factorial	The registered cells completed, but recovery never engaged; no recovery effect is identified.
V1 trigger incidence	Native schema exposure had a positive contrast on the V1 fixed panel; nominal seeds were ineffective and duplicates are collapsed.
V2 directional replication	On the complete outcome-unseen grid, native schema effects were +23, +13, and +10 points and the registered criterion passed. This is conditional on the fixed checkpoints, renderer, and panel.
Structured validity	Invalid structured envelopes occurred less often per 200 requests with native schema in a post-hoc descriptive analysis. Conditional validity fractions are selection-sensitive; this is not the registered primary.
Quest performance	No V1/V2 result establishes downstream state change or quest improvement.
Training superiority	Unsupported; checkpoints are fixed lineage members, not independent training replicates.

D Reproducibility Contract

The public V2 artifact contains portable paths and no endpoint credentials or author identities. Verification requires the expected artifact-index digest and checks the registration, design receipt, snapshot lock, end-point attestations, seed gates, run envelopes, raw JSONL rows, exact generated CSVs, source-code closure, and analysis. The secondary verifier uses strict JSON parsing, reimplements the primary binary parser-recoverability label, recomputes paired contrasts, checks the complete expected path set, and reproduces the directional and heterogeneity verdicts without calling the producer analyzer. Exact recovered-call details also use the shared frozen canonicalization code.

The reviewed PDF records the expected artifact-index digest `04a26f53ce24fa9578c0e49d55b946321347f9de2a1dd81e0739822d57978562` to bind the manuscript to the exact supplementary artifact. The verified artifact tree is `fede8cb870fe8790c5252fd8594f54cdc0505a81ce801f9baed0d036246ecdba`. The anonymous supplement omits the searchable Git revision and the identity-bearing full model lock; those provenance tiers are verified after deanonymization. Internal self-hashes alone are not presented as prior public timestamping.

Historical evidence has a lower grade. Immutable inventories bind recovered directories and a reviewer-facing projection regenerates the nine July scores, but complete transcripts, historical launch receipts, and some executed inputs are not public. Prospective protections do not retroactively upgrade those runs.

E Next Mechanism Tests

The most direct next experiment is an intervention on the routing contract: repair the renderer or endpoint parser, rerun the identical registered state panel, and test whether content-only recoverable calls move into a schema-valid structured channel. A controlled local executor should then report four separate stages: parser acceptance, environment acceptance, immediate state change, and downstream task utility. Replication on an independent state pool, renderer, and open model family is required for a general tool-routing claim.

For the separate training hypothesis, fresh students must begin from the same checkpoint and cross state source with matched histories, strong reset/prefix baselines, independent training seeds, and nested canonical-start evaluation worlds. Results from that study should not be pooled with the fixed-checkpoint interface grid.